

9/13/26

ITSE-2309-005

Lab 2

Lab 2 – SQL SELECT

10 questions • 2 points each • 20 points total

Lab Instructions

In this lab, you will write and execute SQL SELECT statements using the JustLee Books database.

There are 10 questions. Each question is worth 2 points, for a total of 20 points.

What You Need to Do

Complete all 10 questions in order. For every question, you must:

1. Write the SQL query that answers the question.
2. Execute and test your query against the JustLee Books database.
3. Check your results to make sure the query produces the requested output.
4. Copy and paste your SQL code into your submission document.
5. Take a screenshot showing both your executed query and its results.
6. Place the SQL code and screenshot directly under the corresponding question number.

How to Organize Your Submission

Submit ONE document containing all 10 questions.

Organize your work in this exact order:

1.

SQL Code:

```
SELECT TITLE,CATEGORY,PUBDATE AS "Publication Date",RETAIL AS "Retail Price"  
FROM BOOKS  
ORDER BY TITLE;
```

Execution/Results:

	TITLE	CATEGORY	PUBLICATION DATE	RETAIL PRICE
1	BIG BEAR AND LITTL	CHILDREN	11/8/2005, 12:00:00	8.95
2	BODYBUILD IN 10 MI	FITNESS	1/21/2005, 12:00:00	30.95
3	BUILDING A CAR WIT	CHILDREN	3/18/2006, 12:00:00	59.95
4	COOKING WITH MUS	COOKING	2/28/2004, 12:00:00	19.95
5	DATABASE IMPLEME	COMPUTER	6/4/2003, 12:00:00	55.95
6	E-BUSINESS THE EA'	COMPUTER	3/1/2006, 12:00:00	54.5
7	HANDCRANKED COM	COMPUTER	1/21/2005, 12:00:00	25
8	HOLY GRAIL OF ORA	COMPUTER	12/31/2005, 12:00:00	75.95
9	HOW TO GET FASTEI	SELF HELP	11/11/2006, 12:00:00	29.95
10	HOW TO MANAGE TI	BUSINESS	5/9/2003, 12:00:00	31.95
11	PAINLESS CHILD-RE	FAMILY LIFE	7/17/2004, 12:00:00	89.95
12	REVENGE OF MICKE'	FAMILY LIFE	12/14/2005, 12:00:00	22
13	SHORTEST POEMS	LITERATURE	5/1/2005, 12:00:00	39.95
14	THE WOK WAY TO CO	COOKING	9/11/2004, 12:00:00	28.75

2.

SQL Code:

```
SELECT CUSTOMER#,LASTNAME,FIRSTNAME,CITY,STATE
FROM CUSTOMERS
WHERE STATE = 'TX'
ORDER BY LASTNAME, FIRSTNAME;
```

Execution/Result:

	CUSTOMER#	LASTNAME	FIRSTNAME	CITY	STATE
1	1007	GIANA	TAMMY	AUSTIN	TX

3.**SQL Code:**

```
SELECT DISTINCT CATEGORY
FROM BOOKS
ORDER BY CATEGORY;
```

Execution/Result:

	CATEGORY
1	BUSINESS 
2	CHILDREN
3	COMPUTER
4	COOKING
5	FAMILY LIFE
6	FITNESS
7	LITERATURE
8	SELF HELP

4.

SQL Code:

```
SELECT LNAME || ', ' || FNAME AS "Author Name"  
FROM AUTHOR  
ORDER BY LNAME;
```

Execution/Result:

	AUTHOR NAME
1	ADAMS, JUAN
2	AUSTIN, JAMES
3	BAKER, JACK
4	FIELDS, OSCAR
5	JONES, JANICE
6	KZOSCHSKY, TAMARA
7	MARTINEZ, SHEILA
8	PETERSON, TINA
9	PORTER, LISA
10	ROBINSON, ROBERT
11	SMITH, SAM
12	WHITE, WILLIAM
13	WHITE, LISA
14	WILKINSON, AN 

5.

SQL Code:

```
SELECT ORDER#,ITEM#,ISBN, QUANTITY,PAIDEACH,QUANTITY*PAIDEACH AS "Item  
Total"  
FROM ORDERITEMS  
WHERE QUANTITY > 1  
ORDER BY ORDER#;
```

Execution/Result:

	ORDER#	ITEM#	ISBN	QUANTITY	PAIDEACH	ITEM TOTAL
1	1002	1	8843172113	2	55.95	111.9
2	1004	1	2491748320	2	85.45	170.9
3	1007	1	3957136468	3	72.15	216.45
4	1008	1	3437212490	2	19.95	39.9
5	1012	2	1915762492	2	25	50
6	1014	1	0401140733	2	22	44
7	1017	1	8117949391	2	8.95	17.9

6.

SQL Code:

```
SELECT TITLE,NAME AS "Publisher Name",CATEGORY  
FROM BOOKS JOIN PUBLISHER USING(PUBID)  
ORDER BY NAME,TITLE;
```

Execution/Result:

	TITLE		PUBLISHER NAME	CATEGORY
1	DATABASE IMPLEMENTATION	👁	AMERICAN PUBLISHING	COMPUTER
2	HANDCRANKED COMPUTERS		AMERICAN PUBLISHING	COMPUTER
3	HOLY GRAIL OF ORACLE		AMERICAN PUBLISHING	COMPUTER
4	HOW TO MANAGE THE MANAGER		PRINTING IS US	BUSINESS
5	REVENGE OF MICKEY		PRINTING IS US	FAMILY LIFE
6	BUILDING A CAR WITH TOOTHPICKS		PUBLISH OUR WAY	CHILDREN
7	E-BUSINESS THE EASY WAY		PUBLISH OUR WAY	COMPUTER
8	BODYBUILD IN 10 MINUTES A DAY		READING MATERIALS INC.	FITNESS
9	COOKING WITH MUSHROOMS		READING MATERIALS INC.	COOKING
10	HOW TO GET FASTER PIZZA		READING MATERIALS INC.	SELF HELP
11	THE WOK WAY TO COOK		READING MATERIALS INC.	COOKING
12	BIG BEAR AND LITTLE DOVE		REED-N-RITE	CHILDREN
13	PAINLESS CHILD-REARING		REED-N-RITE	FAMILY LIFE
14	SHORTEST POEMS		REED-N-RITE	LITERATURE

7.

SQL Code:

```
SELECT ORDER#,ORDERDATE,CUSTOMER#,FIRSTNAME,LASTNAME
FROM ORDERS JOIN CUSTOMERS USING(CUSTOMER#)
WHERE SHIPSTATE = 'FL'
ORDER BY ORDERDATE;
```

Execution/Result:

	ORDER#	ORDERDATE	CUSTOMER#	FIRSTNAME	LASTNAME
1	1003	4/1/2009, 12:00:00	1001	BONITA	MORALES
2	1006	4/1/2009, 12:00:00	1003	LEILA	SMITH
3	1016	4/4/2009, 12:00:00	1003	LEILA	SMITH
4	1017	4/4/2009, 12:00:00	1015	STEVE	SHELL
5	1018	4/5/2009, 12:00:00	1001	BONITA	MORALES

8.

SQL Code:

```
SELECT ORDER#,TITLE,QUANTITY,PAIDEACH,QUANTITY*PAIDEACH AS "Item Total"
FROM BOOKS NATURAL JOIN ORDERITEMS
WHERE QUANTITY*PAIDEACH > 30
ORDER BY QUANTITY*PAIDEACH DESC;
```

Execution/Result:

	ORDER#	TITLE	QUANTITY	PAIDEACH	ITEM TOTAL
1	1007	HOLY GRAIL OF ORA	3	72.15	216.45
2	1004	PAINLESS CHILD-RE	2	85.45	170.9
3	1002	DATABASE IMPLEME	2	55.95	111.9
4	1001	PAINLESS CHILD-RE	1	85.45	85.45
5	1016	PAINLESS CHILD-RE	1	85.45	85.45
6	1012	PAINLESS CHILD-RE	1	85.45	85.45
7	1011	PAINLESS CHILD-RE	1	85.45	85.45
8	1013	DATABASE IMPLEME	1	55.95	55.95
9	1018	DATABASE IMPLEME	1	55.95	55.95
10	1003	DATABASE IMPLEME	1	55.95	55.95
11	1010	DATABASE IMPLEME	1	55.95	55.95
12	1007	DATABASE IMPLEME	1	55.95	55.95
13	1007	E-BUSINESS THE EA	1	54.5	54.5
14	1006	E-BUSINESS THE EA	1	54.5	54.5
15	1012	HANDCRANKED COM	2	25	50
16	1014	REVENGE OF MICKE	2	22	44
17	1005	SHORTEST POEMS	1	39.95	39.95
18	1008	COOKING WITH MUS	2	19.95	39.9
19	1001	HOW TO MANAGE TI	1	31.95	31.95
20	1003	BODYBUILD IN 10 MI	1	30.95	30.95

9.

SQL Code:

```
SELECT TITLE,CATEGORY,FNAME,LNAME
FROM BOOKS JOIN BOOKAUTHOR USING(ISBN) JOIN AUTHOR USING(AUTHORID)
WHERE CATEGORY = 'COMPUTER'
ORDER BY TITLE,LNAME;
```

Execution/Result:

Query result Script output DBMS output Explain Plan SQL history					
  Download ▾ Execution time: 0.017 seconds					
	TITLE		CATEGORY	FNAME	LNAME
1	DATABASE IMPLEMENTATION		COMPUTER	JUAN	ADAMS
2	DATABASE IMPLEMENTATION		COMPUTER	JAMES	AUSTIN
3	DATABASE IMPLEMENTATION		COMPUTER	TINA	PETERSON
4	E-BUSINESS THE EASY WAY		COMPUTER	JANICE	JONES
5	HANDCRANKED COMPUTERS		COMPUTER	WILLIAM	WHITE
6	HANDCRANKED COMPUTERS		COMPUTER	LISA	WHITE
7	HOLY GRAIL OF ORACLE		COMPUTER	JAMES	AUSTIN

10.

SQL Code:

```
SELECT
CUSTOMER#,FIRSTNAME,LASTNAME,ORDER#,ORDERDATE,TITLE,QUANTITY,PAIDEA
CH,QUANTITY*PAIDEACH AS "Item Total"

FROM CUSTOMERS NATURAL JOIN ORDERS NATURAL JOIN ORDERITEMS NATURAL
JOIN BOOKS

WHERE PAIDEACH > 25

ORDER BY LASTNAME,ORDER#,TITLE;
```

Execution/Result:

Query result Script output DBMS output Explain Plan SQL history									
  Download Execution time: 0.013 seconds									
	CUSTOMER#	FIRSTNAME	LASTNAME	ORDER#	ORDERDATE	TITLE	QUANTITY	PAIDEACH	ITEM TOTAL
1	1020	KENNETH	FALAH	1004	4/1/2009, 12:00:00 AM	PAINLESS CHILD-REARING	2	85.45	170.9
2	1007	TAMMY	GIANA	1007	4/2/2009, 12:00:00 AM	DATABASE IMPLEMENTATION	1	55.95	55.95
3	1007	TAMMY	GIANA	1007	4/2/2009, 12:00:00 AM	E-BUSINESS THE EASY WAY	1	54.5	54.5
4	1007	TAMMY	GIANA	1007	4/2/2009, 12:00:00 AM	HOLY GRAIL OF ORACLE	3	72.15	216.45
5	1014	JASMINE	LEE	1013	4/3/2009, 12:00:00 AM	DATABASE IMPLEMENTATION	1	55.95	55.95
6	1010	JAKE	LUCAS	1001	3/31/2009, 12:00:00 AM	HOW TO MANAGE THE MANAGER	1	31.95	31.95
7	1010	JAKE	LUCAS	1001	3/31/2009, 12:00:00 AM	PAINLESS CHILD-REARING	1	85.45	85.45
8	1010	JAKE	LUCAS	1011	4/3/2009, 12:00:00 AM	PAINLESS CHILD-REARING	1	85.45	85.45
9	1011	REESE	MCGOVERN	1002	3/31/2009, 12:00:00 AM	DATABASE IMPLEMENTATION	2	55.95	111.9
10	1018	GREG	MONTIASA	1005	4/1/2009, 12:00:00 AM	SHORTEST POEMS	1	39.95	39.95
11	1001	BONITA	MORALES	1003	4/1/2009, 12:00:00 AM	BODYBUILD IN 10 MINUTES A DAY	1	30.95	30.95
12	1001	BONITA	MORALES	1003	4/1/2009, 12:00:00 AM	DATABASE IMPLEMENTATION	1	55.95	55.95
13	1001	BONITA	MORALES	1018	4/5/2009, 12:00:00 AM	DATABASE IMPLEMENTATION	1	55.95	55.95
14	1017	BECCA	NELSON	1012	4/3/2009, 12:00:00 AM	PAINLESS CHILD-REARING	1	85.45	85.45
15	1003	LEILA	SMITH	1006	4/1/2009, 12:00:00 AM	E-BUSINESS THE EASY WAY	1	54.5	54.5
16	1019	JENNIFER	SMITH	1010	4/3/2009, 12:00:00 AM	DATABASE IMPLEMENTATION	1	55.95	55.95
17	1003	LEILA	SMITH	1016	4/4/2009, 12:00:00 AM	PAINLESS CHILD-REARING	1	85.45	85.45

Your screenshots must be clear and readable and must show the query execution and results. Before submitting, make sure that every question has both the SQL code and a screenshot of its execution/results.

Working with the Database

To determine the exact names of columns used in the exercises, refer to the tables in the JustLee Books database. You can also examine the structure of a table by using:

```
DESCRIBE tablename;
```

Replace tablename with the name of the table you want to examine.

Important: Execute and test every query yourself. Do not assume that your SQL is correct simply because you wrote it. Check the actual results.

Absolutely no AI may be used to complete this assignment. The purpose of this lab is for you to learn how to write, execute, test, and troubleshoot SQL queries yourself.

Questions

1. Using the BOOKS table, write a query that displays the following information for each book:

- Book title
- Category
- Publication date
- Retail price

Use column aliases so that the publication date and retail price are displayed with the headings "Publication Date" and "Retail Price." Sort the results alphabetically by book title.

2. Using the CUSTOMERS table, write a query that displays the following information for customers who live in Texas:

- Customer number
- First name
- Last name
- City
- State

Display the columns in the order listed above.

Sort the results by last name and then by first name.

3. Using the BOOKS table, write a query that displays all book categories represented in the current inventory. Each category should appear only once in the results. Sort the categories alphabetically.

4. Using the AUTHOR table, write a query that displays each author's last name and first name together in one column.
Format each author's name as:
Last Name, First Name
There must be a comma followed by a blank space between the last and first names.
Use a column alias so that the combined name is displayed with the heading "Author Name."
Sort the results by the author's last name.

5. Using the ORDERITEMS table, write a query that displays the following information for each qualifying order item:
 - Order number
 - Item number
 - ISBN
 - Quantity
 - Amount paid for each itemCreate a calculated column that determines the item total by multiplying Quantity by Paid Each.
Use a column alias so that the calculated column is displayed with the heading "Item Total."
Include only order items with a quantity greater than 1. Sort the results by order number.

6. Using the BOOKS and PUBLISHER tables, write a query that displays the following information for each book:
 - Book title
 - Publisher name
 - CategoryUse a column alias so that the publisher name is displayed with the heading "Publisher Name."
Join the tables using the publisher ID. Sort the results by publisher name and then by book title.

7. Using the ORDERS and CUSTOMERS tables, write a query that displays orders placed by customers who live in Florida.
Display the following columns in the order listed:
 - Order number
 - Order date

- Customer number
- First name
- Last name

Join the tables using the customer number.

Sort the results by order date.

8. Using the ORDERITEMS and BOOKS tables, write a query that displays the following information:

- Order number
- Book title
- Quantity
- Amount paid for each item
- Calculated item total

Calculate the item total by multiplying Quantity by Paid Each.

Use a column alias so that the calculated column is displayed with the heading "Item Total."

Include only records with an item total greater than \$30.

Sort the results from the highest item total to the lowest.

9. Using the BOOKS, BOOKAUTHOR, and AUTHOR tables, write a query that displays books in the COMPUTER category and their authors.

Display the following columns in the order listed:

- Book title
- Category
- Author's first name
- Author's last name

Join the tables using the ISBN and author ID.

Sort the results by book title and then by the author's last name.

10. Using the CUSTOMERS, ORDERS, ORDERITEMS, and BOOKS tables, write a query that displays detailed customer, order, and book information.

Display the following columns in the order listed:

- Customer number
- Customer first name
- Customer last name
- Order number
- Order date
- Book title
- Quantity
- Amount paid for each item
- Calculated item total

Calculate the item total by multiplying Quantity by Paid Each.

Use a column alias so that the calculated column is displayed with the heading "Item Total."

Include only books for which Paid Each is greater than \$25.

Sort the results by customer last name, order number, and book title.

Submission Checklist

Before submitting your lab, verify that:

- You completed all 10 questions.
- Every question contains your SQL code.
- Every question contains a screenshot showing the executed query and its results.
- You executed and tested every query.
- Your screenshots are clear and readable.
- Your questions are clearly numbered 1–10 and presented in order.
- All work is contained in ONE document.
- You reviewed your work for completeness and correctness.

Refer to the lab guidelines and check that all your work is present and correct before submitting.

Submit your completed document using the Lab 2 assignment link in Blackboard.